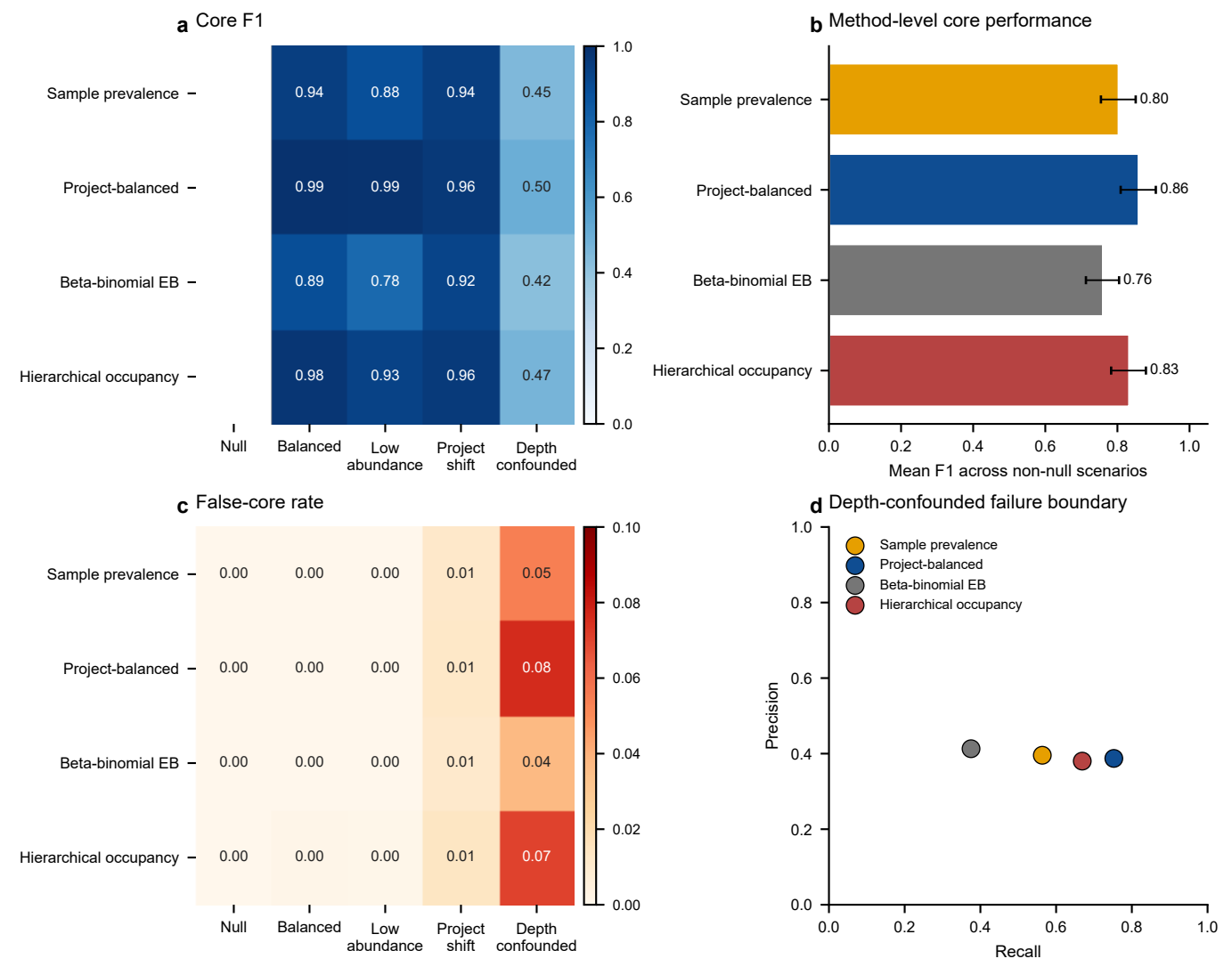
a Core F1
Sample prevalence
Project-balanced
Beta-binomial EB
Hierarchical occupancy
0.94 0.88 0.94 0.45
0.99 0.99 0.96 0.50
0.89 0.78 0.92 0.42
0.98 0.93 0.96 0.47
Null
Balanced
Low abundance
Project shift
Depth confounded
1.0 0.8 0.6 0.4 0.2 0.0
b Method-level core performance
0.80
0.86
0.76
0.83
0.0 0.2 0.4 0.6 0.8 1.0
Mean F1 across non-null scenarios
c False-core rate
0.00 0.00 0.00 0.01 0.05
0.00 0.00 0.00 0.01 0.08
0.00 0.00 0.00 0.01 0.04
0.00 0.00 0.00 0.01 0.07
0.10 0.08 0.06 0.04 0.02 0.00
d Depth-confounded failure boundary
Sample prevalence
Project-balanced
Beta-binomial EB
Hierarchical occupancy
Precision
Recall
0.0 0.2 0.4 0.6 0.8 1.0